

AI Prompt Engineering Assignment

Task 1. AI Prompt

Role: You are a Senior Business Data Analyst with over 8 years of experience in retail analytics, and business performance improvement.

Context: A retail company operating 30 stores has experienced declining performance over the past 18 months. The company has provided a dataset containing approximately 50,000 records covering store performance, product sales, customer transactions, and loyalty programme activities. Available columns include Transaction ID, Customer ID, Date, Gender, Age, Product Category, Quantity, Price per Unit, and Total Amount.

Task: Generate three high-impact business questions that can help identify the causes of declining business performance and uncover opportunities for improvement.

Format: For each business question provide:

1. Business Question
2. Possible Cause Being Investigated
3. Opportunity for Improvement
4. Business Value to the CEO
5. Relevant Data Columns

Constraints:

- Questions must be specific and measurable.
- Avoid technical or statistical jargon.
- Focus on actionable business insights.

Example:

Business Question:

Has product visibility affected sales performance across stores?

Possible Cause Being Investigated:

Low product visibility may reduce customer purchases and overall revenue.

Opportunity for Improvement:

Improve product placement and promotional displays.

Business Value to the CEO:

Helps determine whether merchandising strategies are contributing to declining sales.

Relevant Data Columns:

Product Category, Quantity, Price per Unit, Total Amount, Date.

→ Why did I structure the prompt the way I did ?

I structured the prompt this way because I wanted the AI to have a clear understanding of the company's situation, the available data, and the specific outcome required. By including the role, context, task, format, constraints, and example, I was able to reduce ambiguity and encourage more focused and business-relevant responses. This makes the output more useful for supporting decision-making and identifying improvement opportunities.

Task 2. STAKEHOLDER BRIEF

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Strategic Performance Review: 30-Store Retail Chain

Prepared by: Senior Business Data Analyst · Period: Last 18 months · Records: ~50,000

1. SITUATION SUMMARY: The business has recorded consistent performance decline over the past 18 months across its 30-store network. Early data review indicates the deterioration is unevenly concentrated in specific stores, categories, and customer segments — suggesting structural rather than market-wide causes. A targeted analytical review of the available transaction data can identify the precise sources of revenue loss and surface actionable recovery options before further value erosion occurs.

2. THREE KEY QUESTIONS

Q1 — Store Performance

Which stores are generating the lowest revenue per transaction over the last 6 months, and are those stores declining month-on-month?

Investigates: Structural underperformance vs. short-term dips · Columns: Store ID, Date, Total Amount

Q2 — Product Mix

Which product categories drive 80% of revenue in the top-performing stores but rank outside the top 5 in struggling stores?

Investigates: Misaligned product strategy · Columns: Product Category, Quantity, Price per Unit, Total Amount

Q3 — Customer Retention

Which customers purchased consistently for 6+ months but have not transacted in the last 90 days — and how much revenue do they represent?

Investigates: Lapsed high-value customers · Columns: Customer ID, Date, Total Amount, Loyalty Points

3. DECISIONS THESE QUESTIONS WILL INFORM



Store closures: Q1 identifies which stores have no recovery trajectory, providing a defensible shortlist for closure vs. intervention.



Product prioritisation: Q2 reveals which categories to push into underperforming stores as a low-cost performance lever before any closure decision.



Customer win-back: Q3 quantifies lapsed revenue and enables a targeted, ROI-justified re-engagement campaign rather than broad discounting.

4. DESCRIPTION OF AVAILABLE DATA

Approximately 50,000 transaction records spanning 18 months across 30 stores. The dataset covers customer behaviour, product performance, and loyalty programme activity. Available columns include:

◆ Transaction ID ◆ Customer ID ◆ Date ◆ Gender ◆ Age ◆ Product Category ◆ Quantity
◆ Price per Unit ◆ Total Amount ◆ Loyalty Points